

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
Student Name:	P. MUNI KARTHIK	Reg. No.:	192425061
Date:		Duration:	60 minutes

Code of Conduct

I P. MUNI KARTHIK / 192425061 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: P. MUNI KARTHIK

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. Identify the Project Vision and Stakeholders

- Define the project's vision, objectives, and identify all key stakeholders involved.

Project Vision

Develop a **Smart Warehouse Management System** that automates inventory management, stock tracking, barcode scanning, supplier management, and report generation to improve warehouse efficiency and reduce manual errors.

Project Objectives

- Manage inventory digitally.
- Track stock movement in real time.
- Reduce inventory errors.
- Generate warehouse reports automatically.
- Improve warehouse productivity.

Stakeholders

Stakeholder	Responsibility
Warehouse Manager	Oversees warehouse operations
Inventory Manager	Monitors stock availability
Warehouse Staff	Adds and removes products
Supplier	Supplies products
Administrator	Manages users and system settings
Customers	Receive products accurately

2. Create Epics and User Stories

- Break down the requirements into Epics and formulate well-structured User Stories using the format:
As a <user>, I want <feature>, so that <benefit>.

Epic 1 – User Management

User Story 1

As an Administrator, I want to create user accounts so that authorized users can access the system.

User Story 2

As an Administrator, I want to assign user roles so that users have appropriate permissions.

Epic 2 – Inventory Management

User Story 3

As a Warehouse Staff member, I want to add new stock so that inventory remains updated.

User Story 4

As a Warehouse Staff member, I want to remove damaged stock so that inventory stays accurate.

User Story 5

As an Inventory Manager, I want to update stock quantities so that inventory information is correct.

Epic 3 – Barcode Management

User Story 6

As a Warehouse Staff member, I want to scan product barcodes so that products can be identified quickly.

Epic 4 – Reports

User Story 7

As a Warehouse Manager, I want to generate inventory reports so that I can monitor warehouse performance.

Epic 5 – Supplier Management

User Story 8

As a Supplier, I want to update delivery information so that warehouse staff know shipment status.

3. Define Acceptance Criteria

- Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.

3. Acceptance Criteria

User Story 3 – Add Stock

Given

The warehouse staff is logged in.

When

The staff enters valid product information and clicks **Add Stock**.

Then

The stock is added successfully and inventory quantity is updated.

User Story 4 – Remove Stock

Given

The product exists in inventory.

When

The warehouse staff removes damaged stock.

Then

Inventory quantity decreases accordingly.

User Story 6 – Scan Barcode

Given

A barcode scanner is connected.

When

The barcode is scanned.

Then

The product details appear instantly.

User Story 7 – Generate Reports

Given

Inventory data exists.

When

The warehouse manager clicks **Generate Report**.

Then

The system generates stock and inventory reports successfully.

4. Prioritize the Product Backlog

- Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.

Priority	Product Backlog Item	Business Value
High	User Login	High
High	Add Stock	High
High	Remove Stock	High
High	Barcode Scanning	High
High	Update Inventory	High
Medium	Supplier Management	Medium
Medium	Inventory Reports	Medium
Low	User Role Management	Low

5. Plan the Sprint Backlog

- Select high-priority user stories for the first sprint and divide them into actionable development tasks.

User Story	Development Tasks
User Login	Create login page, Validate users, Database connection
Add Stock	Design stock form, Save product details, Update inventory
Remove Stock	Delete stock records, Update quantity
Barcode Scan	Integrate barcode scanner, Display product details
Inventory Update	Modify stock quantity, Save changes

6. Estimate Effort Using Story Points

- Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.

User Story	Story Points	Justification
User Login	3	Simple authentication
Add Stock	5	Database operations
Remove Stock	5	Inventory validation
Barcode Scanning	8	Hardware integration
Inventory Update	5	Database update logic
Generate Reports	8	Data processing and reporting

7. Present the Sprint Goal and Expected Deliverables

- Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Sprint Goal

Develop the core functionality of the **Smart Warehouse Management System** by implementing secure login, inventory management, and barcode scanning.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification	/5
Epics & User Stories	/5
Acceptance Criteria	/5
Product Backlog Prioritization & Sprint Planning	/5
Story Point Estimation, Sprint Goal & Presentation	/5
TOTAL	/25